



BELLA DONA ERP

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PROJECT DEFINITION AND PROBLEM STATEMENT ✨

Current Situation:

- Business Model: Online sales and direct marketing of gothic necklaces and accessories for women.
- Operational Bottleneck: Information is managed 100% manually using Excel spreadsheets.

Main Challenges:

- Lack of Centralization: Separate records for inventory, daily sales, and customers.
- Operational Inefficiency: Time wasted on manual calculations and visual stock verification.
- Risk of Errors: Frequent human error, stock discrepancies, and unrecorded sales.



THE SOLUTION: BELLA ✨ DONA ERP

01

Centralized Management:
Integrates inventory, sales,
customer CRM, and
supplier data into a single
platform.

02

Custom Categorization:
Classification adapted to
gothic accessories
(materials, chain types,
charms, style labels).

03

Process Automation:
Automatic inventory alerts
and receipt generation to
replace tracking in Excel.

MARKET ANALYSIS & ERP FEASIBILITY



Platform / Tool	Cost / Model	Supplier Mod.	HR / Users Mod.	Inventory / Warehouse	Sales & Billing	CRM / Clients
Treinta (Mobile App)	100% Free (Basic Plan)	Basic: Simple contact registry and accounts	None: Single-user access on mobile	Medium: Simple stock tracking and digital	High: Daily sales tracking, digital	Basic: Basic client address book and
Odoo Community	100% Free (Open Source)	High: Full purchasing module and supplier	Basic: Basic employee management and user	High: Multi-warehouse support, traceability,	High: Point of Sale (POS) and integrated	High: Sales pipeline and client follow-up.
Zoho Inventory	Freemium (Free Plan)	Medium: Purchase order management (Limit: 50	None: Limited to 1 admin user only.	Medium: Stock tracking and item	Medium: Basic commercial invoice	Medium: Contact registry linked to sales
Dolibarr ERP	100% Free (Self-Hosted)	High: Vendor module, orders, and purchase	Medium: Employee management,	High: Stock management,	High: Point of Sale (POS), quotes, and	High: Full management of prospects and

FUNCTIONAL ARCHITECTURE (DECOMPOSITION TREE)



Level 1: Inventory Module

Catalog management (creation, editing, photos, gothic material tags) and stock alerts.

Level 2: Billing and Sales Module

Daily transaction recording, digital receipt generation, and order tracking.

Level 3: Customer Management (CRM)

Customer registration, order history, and style preference tracking.

Level 4: Suppliers and Purchasing

Raw material supplier management and purchase order control.

Level 5: Reports and Analytics

Reports on best-selling necklaces, revenue metrics, and inventory turnover analysis.

NON-FUNCTIONAL REQUIREMENTS AND VISUAL IDENTITY



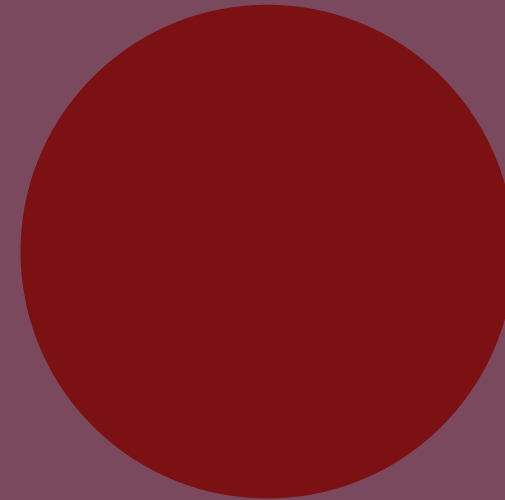
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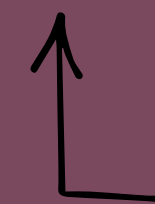
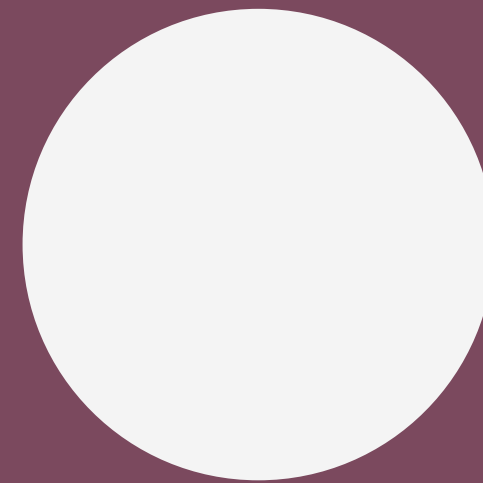
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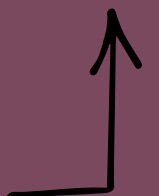
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Secondary
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Indent



Typography and
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NON-FUNCTIONAL REQUIREMENTS AND VISUAL IDENTITY



System Quality Attributes:

01

Usability: Simplified interface requiring fewer than 3 clicks per transaction.

02

Security: Role-based access control (RBAC) to protect financial and vendor data.

03

Availability: Responsive, cloud-hosted web platform accessible 24/7

TRIPLE CONSTRAINT AND SELECTED TECHNOLOGIES



Triple Constraint Framework:

- Scope: Core ERP modules (Inventory, Sales, CRM, Suppliers, Reports). Excludes real-time payment gateway in Phase 1.
- Timeline: Development within the academic semester.
- Cost: Budget of \$0 COP using open-source frameworks and free cloud hosting.

Technology Stack:

- Microsoft web
- C#
- SQL SERVER
- Visual Studio code
- Git hub

USER ROLES



Role	Access Level	Core Functions & Responsibilities
Administrator	Full Access	Manages products, inventory, sales, customers, suppliers, and executive reports.
Salesperson / Sales Operator	Commercial Access	Registers daily transactions, queries product availability, and processes customer orders.
Inventory Manager	Operational Access	Logs stock inflows/outflows, monitors inventory levels, and
Client (If in scope)	Public Access	Browses catalog information and submits purchase requests/orders.



CONCLUSIONS



- Automation: Replace manual Excel tracking with a centralized, error-free web-based system.
- Custom Approach: Designed specifically for gothic jewelry, overcoming the limitations of generic software.
- Zero Cost (\$0): High-value solution developed with open-source technologies and free hosting.
- Scalability: Optimize sales and inventory while ensuring security through user roles.

